Associativity of the Zappa–Szép Product and the Four Cocycle Axioms

Extracted from a machine-checked development

Abstract

We formalise the algebraic equivalence between associativity of the Zappa–Szép product of two monoids and the four Brin/Zappa–Szép cocycle axioms. Given monoids L and G with a left action $\triangleright : G \to L \to L$ and a restriction $\triangleleft : G \to L \to G$, subject to unit-compatibility conditions that make $(1_L, 1_G)$ a two-sided unit, the product

$$(a, g) * (b, h) \;=\; \big(a \cdot (g \triangleright b),\ (g \triangleleft b) \cdot h\big)$$

on $L \times G$ is associative if and only if the four cocycle axioms ZS1–ZS4 hold. The proofs of all stated results have been machine-checked; we record only the mathematical content.

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1 Introduction

The Zappa–Szép product is a generalisation of the semidirect product in which both factors act on each other. Algebraically, one is given two monoids L and G, a left action of G on L, and a restriction of G along elements of L; the product monoid structure on the cartesian product $L \times G$ is then defined by a mixed formula that uses both operations. For this product to be associative, the actions must satisfy four cocycle identities, often called the Brin or Zappa–Szép axioms.

This note records a pure equational proof that, once unit-compatibility is assumed (so that $(1_L, 1_G)$ is a two-sided unit), associativity of the product is equivalent to the four cocycle axioms. The forward direction uses no unit laws. The converse recovers each axiom by specialising associativity at units and reading off one coordinate: the L-coordinate yields ZS1 and ZS2; the G-coordinate yields ZS3 and ZS4.

We work with types as carriers and ordinary functions as operations. Proofs are indicated by the phrase "by lean."

What this note claims. For a matched pair of monoids (two monoids, two mutual actions, and four unit compatibilities),

$$\text{ZS1} \wedge \text{ZS2} \wedge \text{ZS3} \wedge \text{ZS4} \iff \text{associativity of the Zappa–Szép product.}$$

There is nothing more to the associativity law than these four cocycles, and conversely each cocycle is an instance of associativity.

2 Matched pairs of monoids

Definition 2.1 (Matched pair)**.** A **matched pair** of monoids consists of:

1. monoids $(L, \cdot_L, 1_L)$ and $(G, \cdot_G, 1_G)$, each associative and unital;
2. a left action $\triangleright : G \to L \to L$;
3. a restriction $\triangleleft : G \to L \to G$;
4. the four **unit-compatibility** conditions

$$1_G \triangleright a = a, \tag{U1a}$$
$$1_G \triangleleft a = 1_G, \tag{U1b}$$
$$g \triangleright 1_L = 1_L, \tag{U2a}$$
$$g \triangleleft 1_L = g, \tag{U2b}$$

for all $a \in L$ and $g \in G$.

No cocycle axiom is assumed as part of this definition; ZS1–ZS4 are stated separately below.

Remark 2.2. The unit-compatibility conditions are exactly what make $(1_L, 1_G)$ a two-sided unit for the product operation of the next section.

Definition 2.3 (Zappa–Szép product)**.** Let P be a matched pair. The **Zappa–Szép product** is the binary operation on $L \times G$ defined by

$$(a, g) * (b, h) = \big(a \cdot_L (g \triangleright b),\ (g \triangleleft b) \cdot_G h\big). \tag{1}$$

3 The four cocycle axioms

Definition 3.1 (Cocycle axioms)**.** Let P be a matched pair. The four **Brin/Zappa–Szép cocycle axioms** are the following predicates on P.

$$g \triangleright (a \cdot_L b) = (g \triangleright a) \cdot_L \big((g \triangleleft a) \triangleright b\big), \tag{ZS1}$$
$$(g \cdot_G h) \triangleright a = g \triangleright (h \triangleright a), \tag{ZS2}$$
$$(g \cdot_G h) \triangleleft a = \big(g \triangleleft (h \triangleright a)\big) \cdot_G (h \triangleleft a), \tag{ZS3}$$
$$g \triangleleft (a \cdot_L b) = (g \triangleleft a) \triangleleft b. \tag{ZS4}$$

In words: ZS1 says that $\triangleright$ is a crossed homomorphism in its L-argument; ZS2 says that $\triangleright$ is a left action of G; ZS3 says that $\triangleleft$ is a crossed homomorphism in its G-argument; ZS4 says that $\triangleleft$ is a right action of L.

Definition 3.2 (Associativity)**.** The product of Definition 2.3 is **associative** when

$$\big((x * y) * z\big) \;=\; \big(x * (y * z)\big) \tag{2}$$

for all $x, y, z \in L \times G$.

4 Cocycles imply associativity

Theorem 4.1 (Forward direction)**.** *Let P be a matched pair. If ZS1–ZS4 all hold, then the Zappa–Szép product is associative.*

Proof. By lean. □

Remark 4.2 (Idea of the argument)**.** Expand both sides of the associativity equation at a general triple $((a, g), (b, h), (c, k))$. The L-coordinate of the resulting equality follows from reassociating $\cdot_L$, then ZS1 (to distribute the action over a product) and ZS2 (the action law). The G-coordinate follows from ZS3, then reassociating $\cdot_G$, then ZS4. No unit law is used.

5 Associativity implies the cocycles

Theorem 5.1 (Converse direction)**.** *Let P be a matched pair. If the Zappa–Szép product is associative, then ZS1, ZS2, ZS3, and ZS4 all hold.*

Proof. By lean. □

Remark 5.2 (How each axiom is recovered)**.** Each cocycle is an instance of associativity at units.

1. **ZS1.** Specialise at $x = (1_L, g)$, $y = (a, 1_G)$, $z = (b, 1_G)$ and read the L-coordinate; unit laws collapse the remaining factors.
2. **ZS2.** Specialise at $x = (1_L, g)$, $y = (1_L, h)$, $z = (a, 1_G)$ and read the L-coordinate.
3. **ZS3.** The same triple as for ZS2, reading the G-coordinate.
4. **ZS4.** The same triple as for ZS1, reading the G-coordinate.

The unit-compatibility fields of the matched pair are exactly what make these specialisations yield the bare cocycle equations.

6 The equivalence

Theorem 6.1 (Main theorem)**.** *For any matched pair of monoids, the Zappa–Szép product is associative if and only if the four cocycle axioms ZS1–ZS4 hold:*

$$(\mathrm{ZS1} \wedge \mathrm{ZS2} \wedge \mathrm{ZS3} \wedge \mathrm{ZS4}) \iff \mathrm{Assoc.} \tag{3}$$

Proof. By lean. □

7 An example: the direct product

Example 7.1 (Trivial actions on $(\mathbb{N}, +, 0)$)**.** Take $L = G = \mathbb{N}$ with addition as multiplication and 0 as unit, and define the trivial actions

$$g \triangleright a \;=\; a, \qquad g \triangleleft a \;=\; g. \tag{4}$$

These satisfy the four unit-compatibility conditions, so one obtains a matched pair. All four cocycle axioms hold (both sides of each equation reduce to the same expression by the definitions of the actions). Consequently, by Theorem 6.1, the Zappa–Szép product is associative. In this case the product is simply the direct product monoid $(\mathbb{N} \times \mathbb{N},\, +,\, (0, 0))$.

Remark 7.2. The example serves as a sanity check that the matched-pair structure is inhabited and that ZS1–ZS4 are satisfiable.

8 Discussion

What has been established. Associativity of the Zappa–Szép product on a matched pair is equivalent to four equational cocycle axioms. The proof is purely equational: the forward direction is a direct expansion; the converse is specialisation at units. Nothing beyond monoid laws, unit compatibility, and the two mutual actions is used.

Scope. The development treats monoids (not necessarily groups). It does not construct the free or universal matched pair, nor does it address topological, ordered, or higher-categorical refinements of the Zappa–Szép product. The unit laws of the product monoid are not stated as separate theorems; they are the content of the unit-compatibility package built into Definition 2.1.

Machine checking. Every definition and theorem statement above corresponds to a declaration in a machine-checked formalisation. The phrase "by lean" indicates that the proof has been verified by a proof assistant.

Acknowledgements

The mathematical content follows the classical theory of Zappa–Szép products of monoids and the Brin cocycle presentation of the associativity criteria. The presentation here is a prose rendering of a formal development intended for human review of the theorem statements.

References

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